$$N(\log\log N)^{-2^{-2^{s+9}}}$$

elements contains an arithmetic progression of length s.

Typically proofs using ergodic theory are not effective: Theorem 1.5 easily implies a finitistic version of Szemerédi's theorem, which states that for every s and constant $c > 0$ and all sufficiently large $N = N(s, c)$, any subset of $\{1, \dots, N\}$ with at least cN elements contains an arithmetic progression of length s. However, the dependence of N on c is not known by this means, nor is it easily deduced from the proof of Theorem 1.5. Gowers' Theorem, proved by different methods, does give an explicit dependence.

We mention Gowers' Theorem to indicate some of the limitations of ergodic theory. While ergodic methods have many advantages, proving quite general theorems which often have no other proofs, they also have disadvantages, one of them being that they tend to be non-effective.

Subsequent development of the combinatorial and arithmetic ideas by Goldston, Pintz and Yıldırım [118](3) and Gowers, and of the ergodic method by Host and Kra [159] and Ziegler [393], has influenced some arguments of Green and Tao [127] in their proof of the following long-conjectured result. This is a good example of how asking for effective or quantitative versions of existing results can lead to new qualitative theorems.

Theorem (Green and Tao). The set of primes contains arbitrarily long arithmetic progressions.

1.4 Indefinite Quadratic Forms and Oppenheim's Conjecture

Our purpose here is to provide enough background in ergodic theory to quickly reach some understanding of a few deeper results in number theory and combinatorial number theory where ergodic theory has made a contribution. Along the way we will develop a good portion of ergodic theory as well as some other background material. In the rest of this introductory chapter, we mention some more highlights of the many connections between ergodic theory and number theory. The results in this section, and in Sects. 1.5 and 1.6, will not be covered in this book, but we plan to discuss them in a subsequent volume.

The next theorem was conjectured in a weaker form by Oppenheim in 1929 and eventually proved by Margulis in the stronger form stated here in 1986 [247, 250]. In order to state the result, we recall some terminology for quadratic forms.

A *quadratic form* in n variables is a homogeneous polynomial $Q(x_1, \dots, x_n)$ of degree two. Equivalently, a quadratic form is a polynomial Q for which there is a symmetric $n \times n$ matrix A_Q with

$$Q(x_1,\dots,x_n)=(x_1,\dots,x_n)A_Q(x_1,\dots,x_n)^{\mathrm{t}}.$$

Since A_Q is symmetric, there is an orthogonal matrix P for which $P^{\mathrm{t}}A_QP$ is diagonal. This means there is a different coordinate system $y_1,\dots,y_n$ for which

$$Q(x_1,\dots,x_n)=c_1y_1^2+\cdots+c_ny_n^2.$$

The quadratic form is called *non-degenerate* if all the coefficients c_i are non-zero (equivalently, if $\det A_Q\neq 0$), and is called *indefinite* if the coefficients c_i do not all have the same sign. Finally, the quadratic form is said to be *rational* if its coefficients (equivalently, if the entries of the matrix A_Q) are rational*.

Theorem (Margulis). Let Q be an indefinite non-degenerate quadratic form in $n\geqslant 3$ variables that is not a multiple of a rational form. Then $Q(\mathbb{Z}^n)$ is a dense subset of $\mathbb{R}$.

It is easy to see that two of the stated conditions are necessary for the result: if the form Q is definite then the elements of $Q(\mathbb{Z}^n)$ all have the same sign, and if Q is a multiple of a rational form, then $Q(\mathbb{Z}^n)$ lies in a discrete subgroup of $\mathbb{R}$. The assumption that Q is non-degenerate and n is at least 3 are also necessary, though this is less obvious (requiring in particular the notion of badly approximable numbers from the theory of Diophantine approximation, which will be introduced in Sect. 3.3). This shows that the theorem as stated above is in the strongest possible form. Weaker forms of this result have been obtained by other methods, but the full strength of Margulis' Theorem at the moment requires dynamical arguments (for example, ergodic methods).

Proving the theorem involves understanding the behavior of *orbits* for the action of the subgroup $\mathrm{SO}(2,1)\leqslant\mathrm{SL}_3(\mathbb{R})$ on points $x\in\mathrm{SL}_3(\mathbb{Z})\backslash\mathrm{SL}_3(\mathbb{R})$ (the space of right cosets of $\mathrm{SL}_3(\mathbb{Z})$ in $\mathrm{SL}_3(\mathbb{R})$); these may be thought of as sets of the form $x\,\mathrm{SO}(2,1)$. As it turns out (a consequence of Raghunathan's conjectures, discussed briefly in Sect. 1.7), such orbits are either closed subsets of $\mathrm{SL}_3(\mathbb{Z})\backslash\mathrm{SL}_3(\mathbb{R})$ or are dense in $\mathrm{SL}_3(\mathbb{Z})\backslash\mathrm{SL}_3(\mathbb{R})$. Moreover, the former case happens if and only if the point x corresponds in an appropriate sense to a rational quadratic form.

Margulis' Theorem may be viewed as an extension of Example 1.3 to higher degree in the following sense. The statement that every orbit under the map $R_\alpha(t)=t+\alpha\pmod 1$ is dense in $\mathbb{T}$ is equivalent to the statement that if L is a linear form in two variables that is not a multiple of a rational form, then $L(\mathbb{Z}^2)$ is dense in $\mathbb{R}$.

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* Note that the rationality of Q cannot be detected using the coefficients $c_1,\dots,c_n$ after the real coordinate change.